PLANETARY PS

AN INDEPENDENT PLANETARY PS READOUT

The AI build-out meets the physical world

DATE October 14, 2026

COVERS Demand, power, and place

SUMMARY

Expansion needs credible demand, deliverable clean power, and enforceable local terms

DEMAND

Stage commitments as demand becomes real

Data-center use could roughly double by 2030, but forecasts differ and announcements run ahead of delivery.

- IEA's Base Case reaches 945 TWh by 2030; US estimates for 2030 range from 426 to 649 TWh.
- Of 189 GW of gas planned for data centers, 16.9 GW is under construction.

POWER

Show how clean electricity reaches the facility

Gas grows most in IEA's projections, but solar, wind, and batteries are most of what is being built.

- If built, proposed data-center gas plants would emit about 318 Mt of CO_2 a year, a fifth of US power emissions.
- Batteries now carry solar into California's evening peak, and AI workloads have flexed in a trial.

PLACE

Put protections and benefits into the agreement

Water, land, and grid costs fall on host communities, and residents raise water most often.

- PJM's market monitor attributes 38% of its latest capacity charges to data-center load.
- States are adding tariffs, permit pauses, and disclosure rules, mostly from governors and regulators.

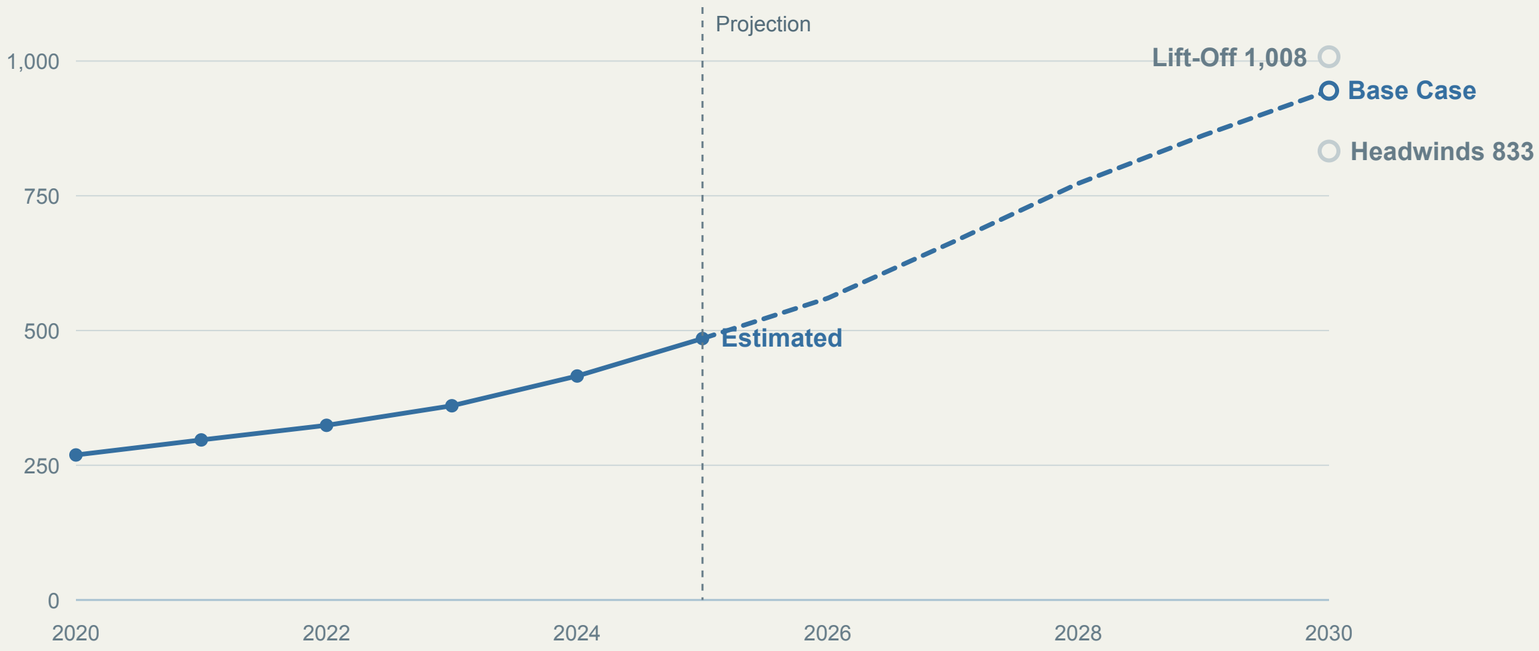
Sources: IEA (2025, 2026); LBNL (2026); Global Energy Monitor (2026); Bloomberg, citing BloombergNEF (2026); EIA (2026); Ember (2026); Monitoring Analytics (2026); Brockovich map. Detail and links on each slide.

Data-center electricity use roughly doubles by 2030 in IEA's Base Case

Use grew 17% in 2025 to about 485 TWh, slightly more than 1.5% of global electricity. The Base Case reaches 945 TWh by 2030.

Global data-center electricity consumption

TWh; IEA estimates to 2025; dots mark other 2030 cases



Source: IEA, Key Questions on Energy and AI (April 2026), chart data and Tables A.1 and A.4.

SO WHAT

Two countries carry most of the growth

- US use rises from 224 TWh in 2025 to 426 TWh in 2030 in the Base Case
- China rises from 117 to 277 TWh over the same years

SCOPE

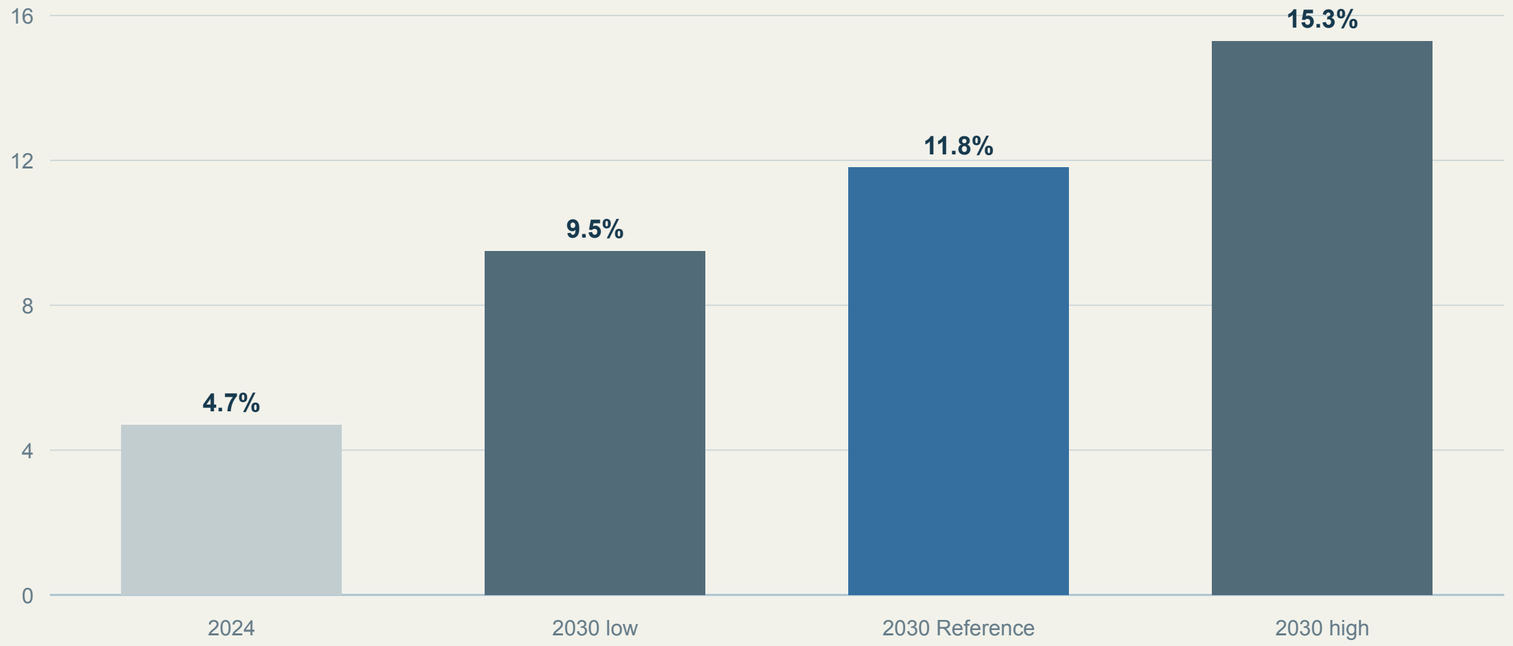
All data centers, not AI alone. The 2030 cases are scenarios, not a probability range.

US data centers could use 9.5–15.3% of the country's electricity by 2030

Their share was 4.7% in 2024. LBNL's 2030 Reference Case is 11.8%, or 649 TWh, more than three times 2024.

Data-center share of US electricity consumption

Percent; LBNL 2025 Update; 2030 low and high cases bound its scenarios



SO WHAT

Estimates differ by model

- IEA's US Base Case is 426 TWh in 2030; LBNL's Reference Case is 649 TWh
- EPRI's 2030 range is 380–790 TWh, or 9% to 17% of US electricity

READ WITH CARE

The low and high cases are scenario bounds, not probabilities.

Sources: LBNL, US Data Center Energy Usage Report: 2025 Update (June 2026), Table 2, national share on NERC's projected denominator; IEA (April 2026), Table A.4; EPRI, Powering Intelligence 2026.

Announcements become usable capacity only when demand and infrastructure arrive

Track each project through connection and load ramp-up. Evidence stages can overlap; requests can duplicate or change.

Announced

A location, capacity, and opening claim.

Secured

Demand, financing, permits, and power arrangements.

Built

Verifiable progress on the facility and its connection.

Energized

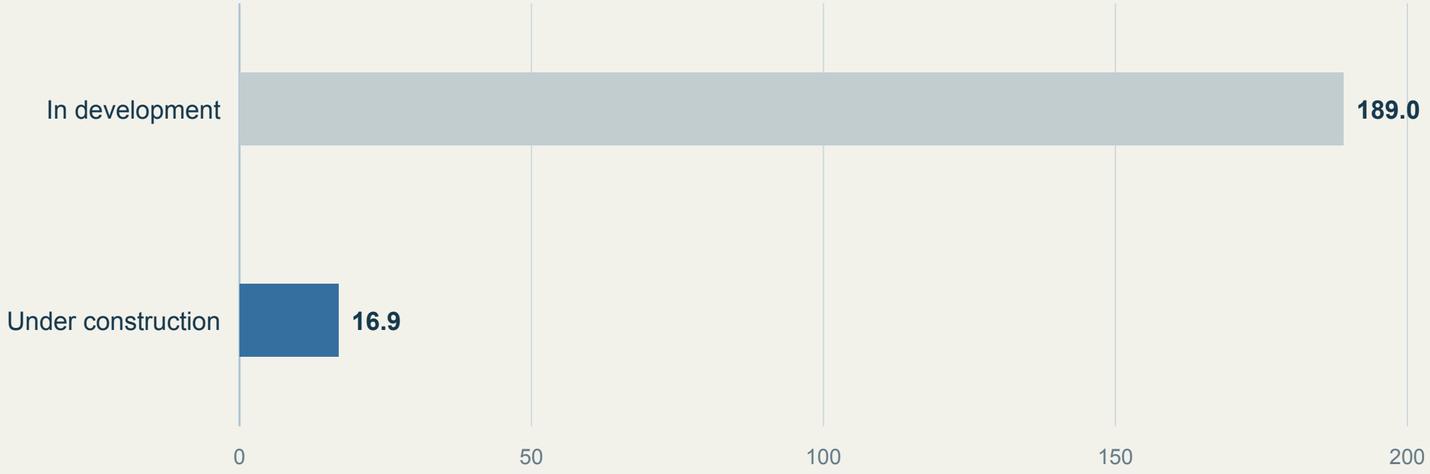
The connection and facility are ready to operate.

Utilized

Equipment and workloads reach their planned load.

A gas example: only part of the pipeline is being built

GW; US gas projects intended to directly power data centers; August 2026



DELIVERY EVIDENCE

Spending also needs context

- Seven cloud companies plan about $713 billion of capex in 2026, up almost three-quarters
- Currence attributes the latest increases to component prices, especially memory

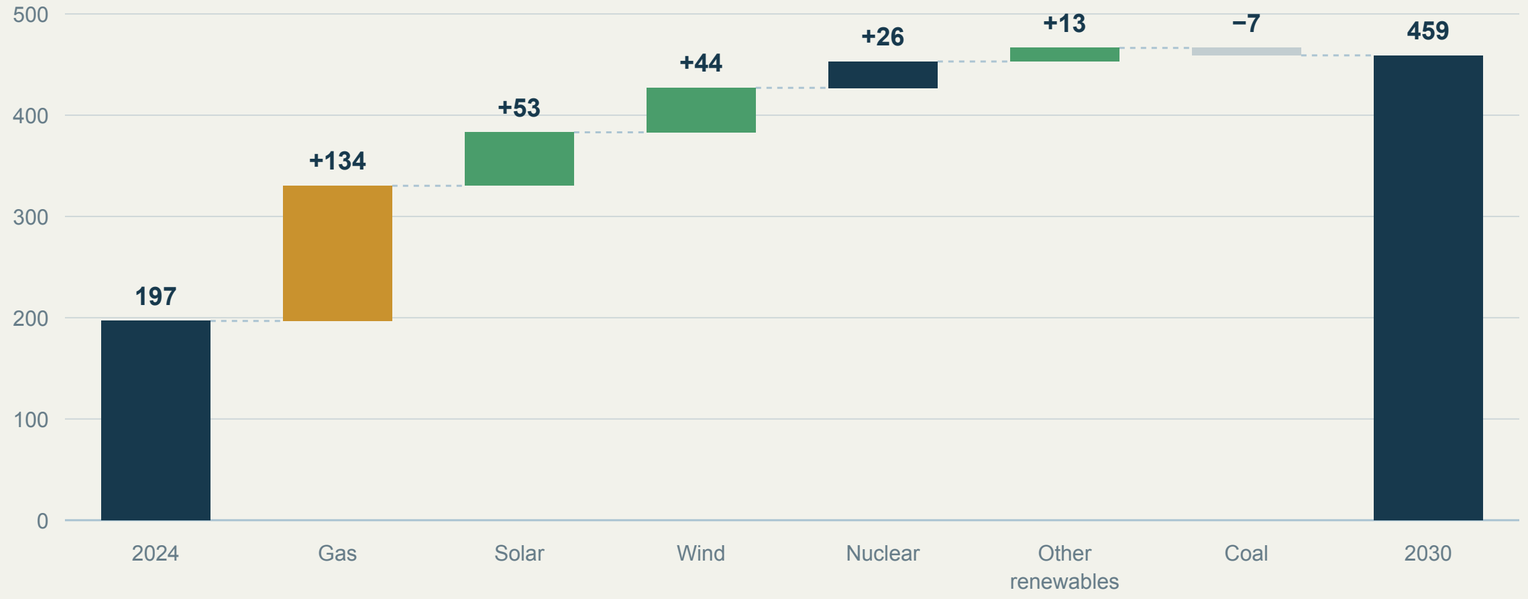
Sources: GEM (August 2026): 189 GW in announced, pre-construction, and construction stages; 16.9 GW is a subset. IEA capex chart; Currence Q3 commentary (August 2026). Stages are a Planetary PS synthesis informed by Currence's methodology.

In IEA's projections, gas is the largest source of new power for US data centers

In IEA's 2025 Base Case, US data-center supply grows from about 197 TWh in 2024 to 459 TWh in 2030. Gas adds 134 TWh, more than solar and wind combined.

Electricity supply for US data centers, 2024 to 2030, by source

TWh, IEA 2025 Base Case



Source: IEA, Energy and AI (April 2025), Base Case, electricity supply for US data centers by source. Step sizes are Planetary PS calculations from IEA chart data.

SO WHAT

Gas grows the most in this scenario

- Gas supply more than doubles, from 86 to 220 TWh
- Coal is the only source that shrinks

EDITION

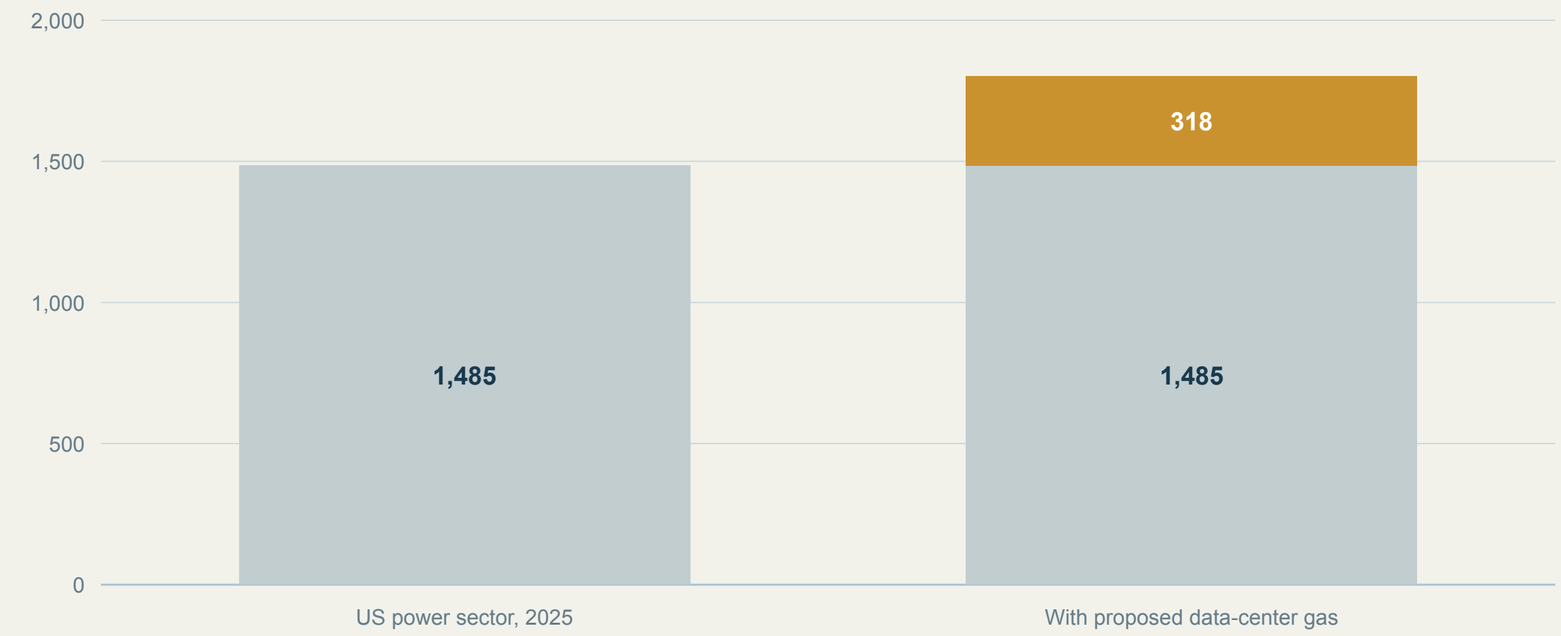
These are IEA's 2025 projections; the April 2026 update does not publish source-level detail for the US.

If built, proposed data-center gas plants would add CO_2 equal to a fifth of US power emissions

BloombergNEF estimates the 99 plants it tracks would emit about 318 million tonnes of CO_2 a year at industry-standard rates.

US power-sector CO_2 emissions

Million tonnes a year; 2025, and with the proposed plants' own emissions added



LOCK-IN

Most of it is still a choice

- Gas plants approved now would run for decades
- On GEM's separate project list, 16.9 GW of 189 GW is under construction

READ WITH CARE

A gross figure: CO_2 from burning the gas, before any generation it displaces and without upstream methane.

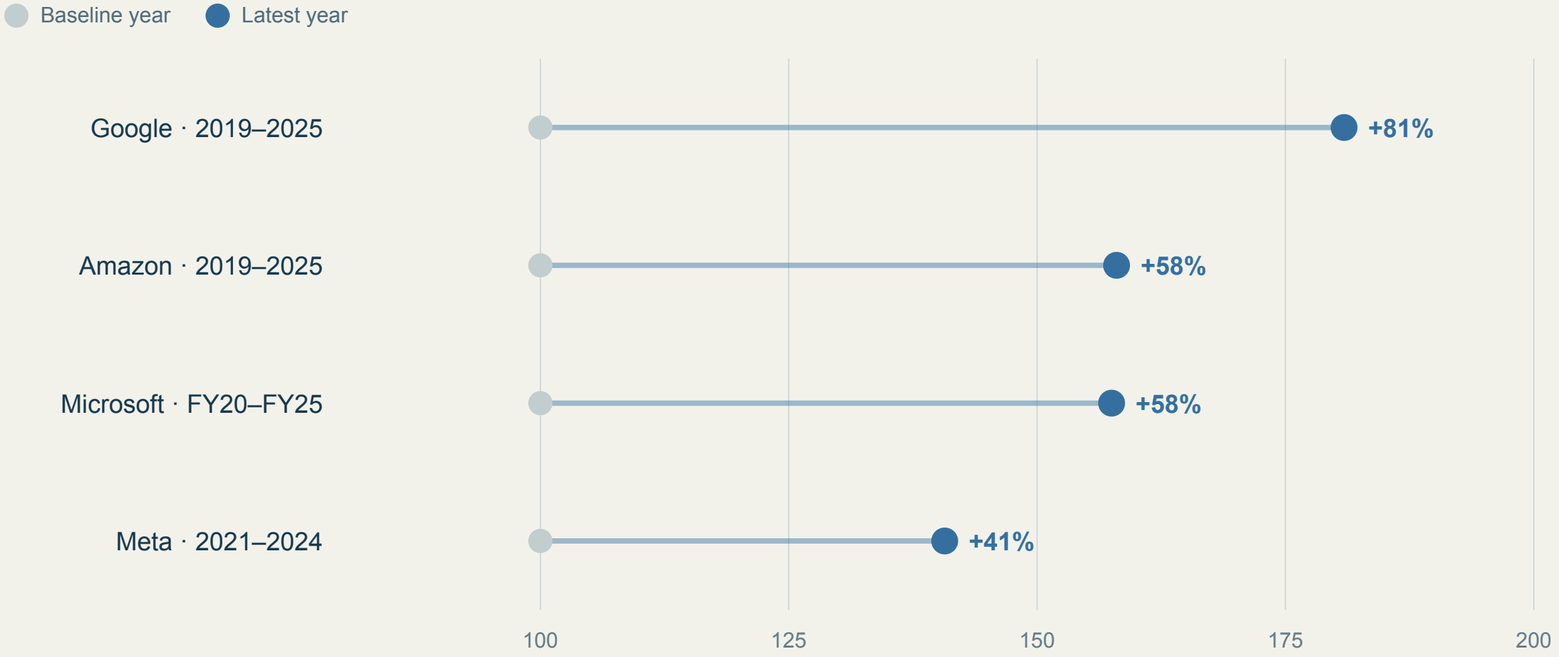
Sources: Bloomberg, "Data center gas plants to boost US power emissions by 20%" (August 18, 2026), citing BloombergNEF and EIA; Global Energy Monitor (August 2026). The two trackers cover different project sets.

The four largest hyperscalers' reported emissions are up 41–81% since their baselines

Each company reports higher totals as its data centers expand, while its climate target points the other way.

Total reported emissions, indexed to each company's baseline year

Baseline year = 100; each company's own baseline, boundary, and target basis; market-based



WHAT THE COMPANIES REPORT

+81%

Google, 2019 to 2025

Google's own figure, on its target basis

+25%

Microsoft, FY24 to FY25

Partly from dropping unbundled certificates

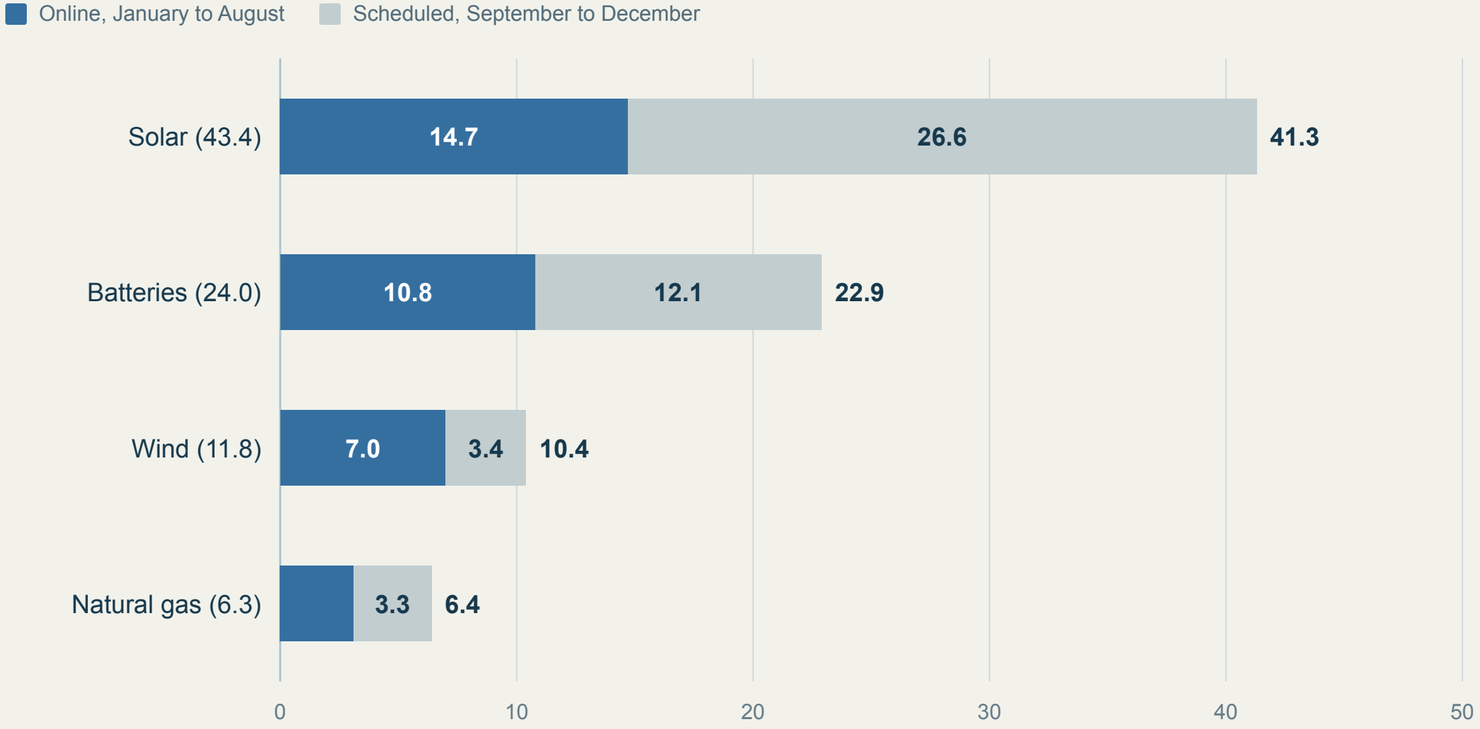
Sources: Microsoft 2026 Environmental Data Fact Sheet; Google 2026 Environmental Report; Amazon 2025 Sustainability Report (whole company; 2019 is its first reported year); Meta 2025 Environmental Data Index. Changes other than Google's are Planetary PS calculations.

Solar, wind, and batteries are arriving close to plan in 2026

36 GW came online from January through August, and 45 GW more is scheduled by December. Together that is 81 GW, against the 86 GW developers planned in February.

US utility-scale capacity coming online in 2026, by technology

GW of nameplate capacity; EIA inventory as of August 2026; February plan in brackets



WHAT IS LEFT

Nearly all of it is already being built

- 18 GW of the 45 GW still scheduled is built and awaiting commercial operation
- Another 27 GW is under construction

READ WITH CARE

Nameplate capacity, not energy delivered. Batteries store power rather than generate it.

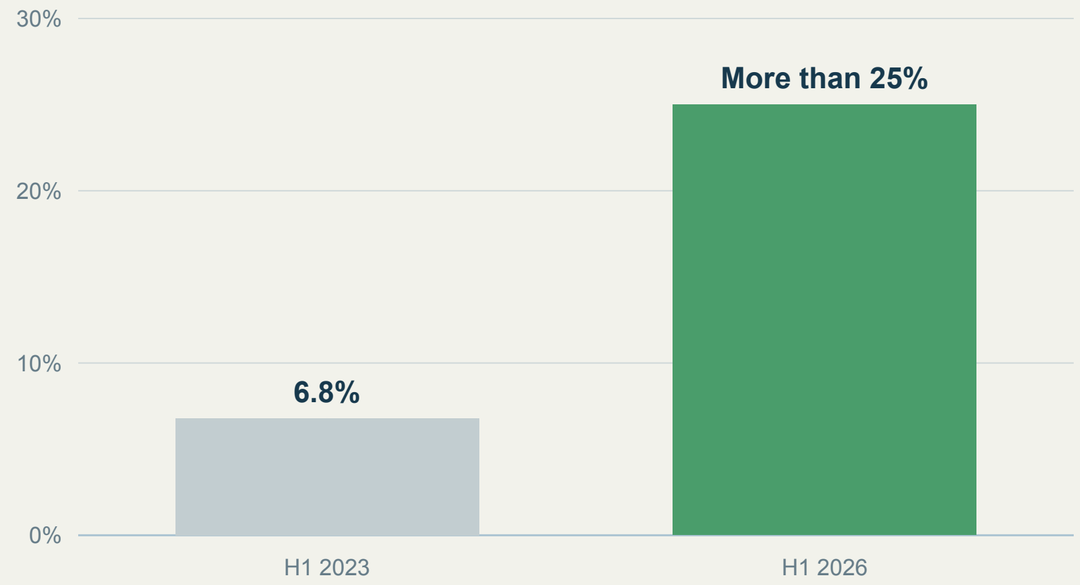
Sources: EIA, Preliminary Monthly Electric Generator Inventory (August 2026 data, released September 24, 2026); February plan: EIA, Today in Energy (February 20, 2026). Sums by technology are Planetary PS calculations; Puerto Rico excluded.

Batteries and flexible demand are stretching clean power into more hours

Solar plus batteries now meet more than a quarter of California’s evening peak. In a Phoenix trial, an AI cluster cut its power 25% for three hours and still met its service guarantees.

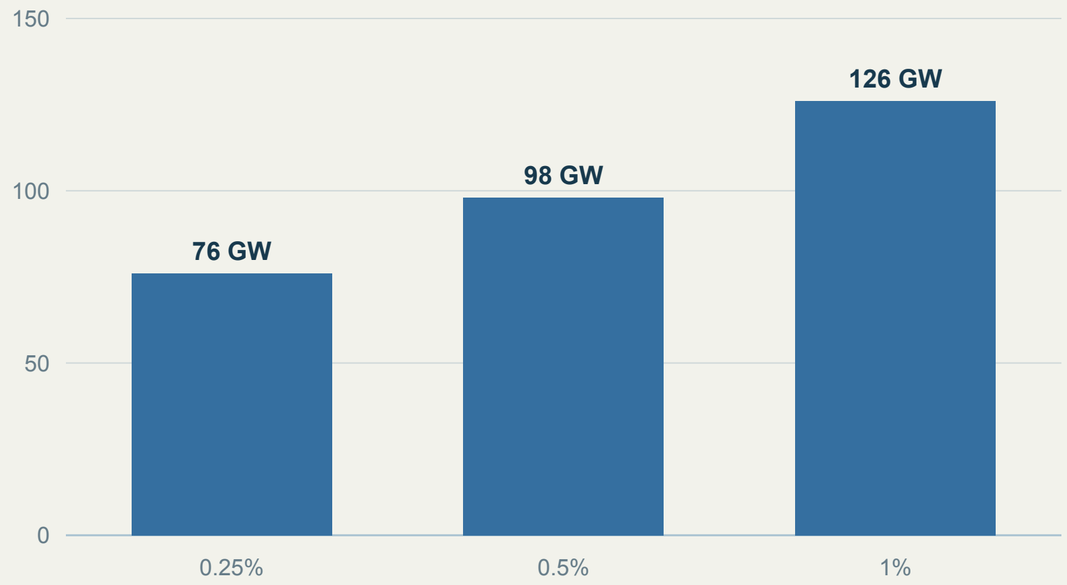
Solar plus batteries, share of California’s evening demand

7–9pm on the average day, first half of each year; CAISO area



New load the existing grid could absorb with flexibility

GW, by average annual curtailment rate; 22 largest balancing authorities



Sources: Ember, Batteries have unlocked the era of anytime solar (August 2026), p. 19; Norris et al., Rethinking Load Growth, Duke Nicholas Institute (February 2025), which does not model transmission limits; Colangelo et al., Phoenix field demonstration (2025), published in Nature Energy. One trial does not show every campus can flex.

A campus changes the places around it and the places that supply its power

Cooling and electricity choices connect local resource demand to wider environmental pressures.

AT THE CAMPUS

Cooling, construction, and backup power

Water sourcing, clearing land, generator emissions, and noise affect people and habitats around the site.

BEYOND THE CAMPUS

The electricity supply carries impacts too

Generating its power can use water and emit pollution in another watershed and community.

Follow the water

Identify source, season, consumption, and withdrawals. Check drought conditions and competing users.

Read the landscape

Assess habitat, wetlands, farmland, and cumulative development before choosing the site.

Count who is exposed

Include neighboring communities, existing pollution, and people affected by supporting power plants.

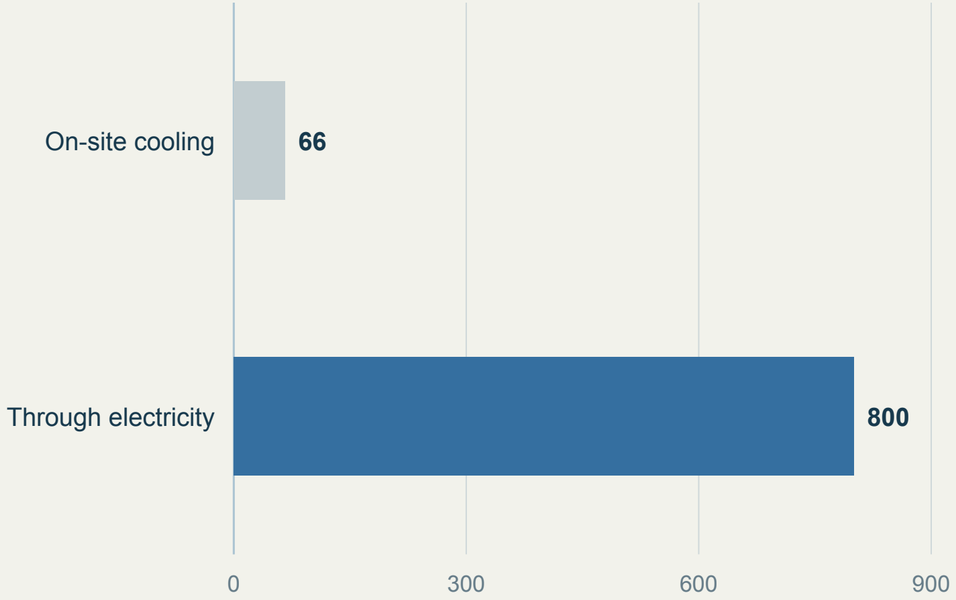
Sources: WRI, From Energy Use to Air Quality, the Many Ways Data Centers Affect US Communities (February 17, 2026); water through electricity: LBNL (December 2024). Pathways summarized by Planetary PS; no site-specific assessment is implied.

Nationally, data centers use far more water through their electricity than on site

US data centers consumed about 66 billion liters on site in 2023 and nearly 800 billion through the power they used. On-site cooling still draws on one local watershed: about two-thirds of US data centers built since 2022 are in water-stressed areas.

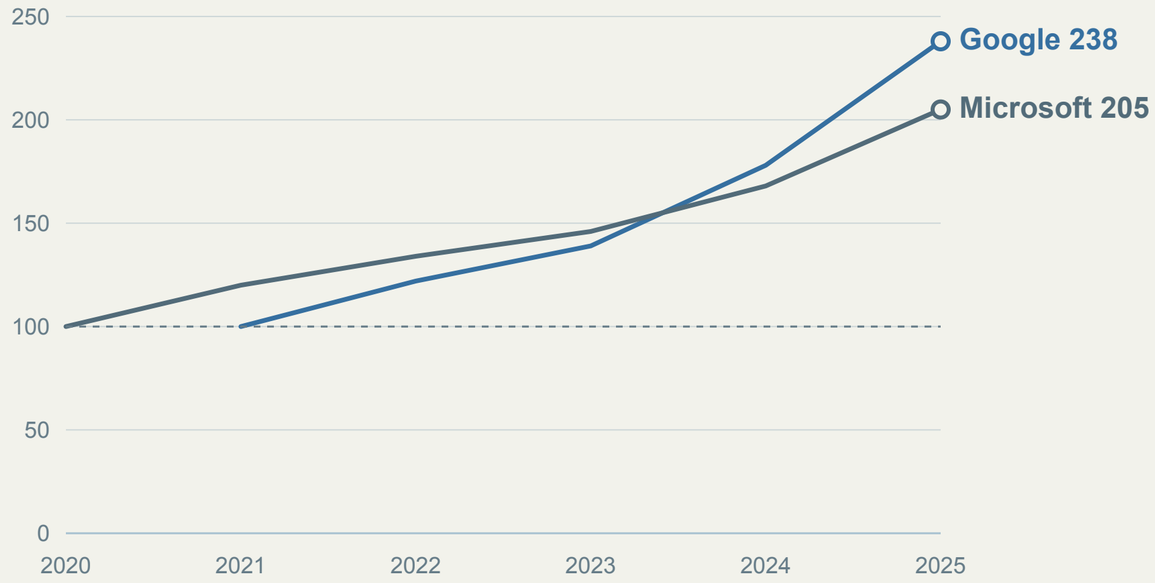
US data-center water consumption, 2023

Billion liters; national totals; indirect is LBNL's "nearly 800"



Company water consumption, indexed

First year shown = 100; Microsoft fiscal years end in June



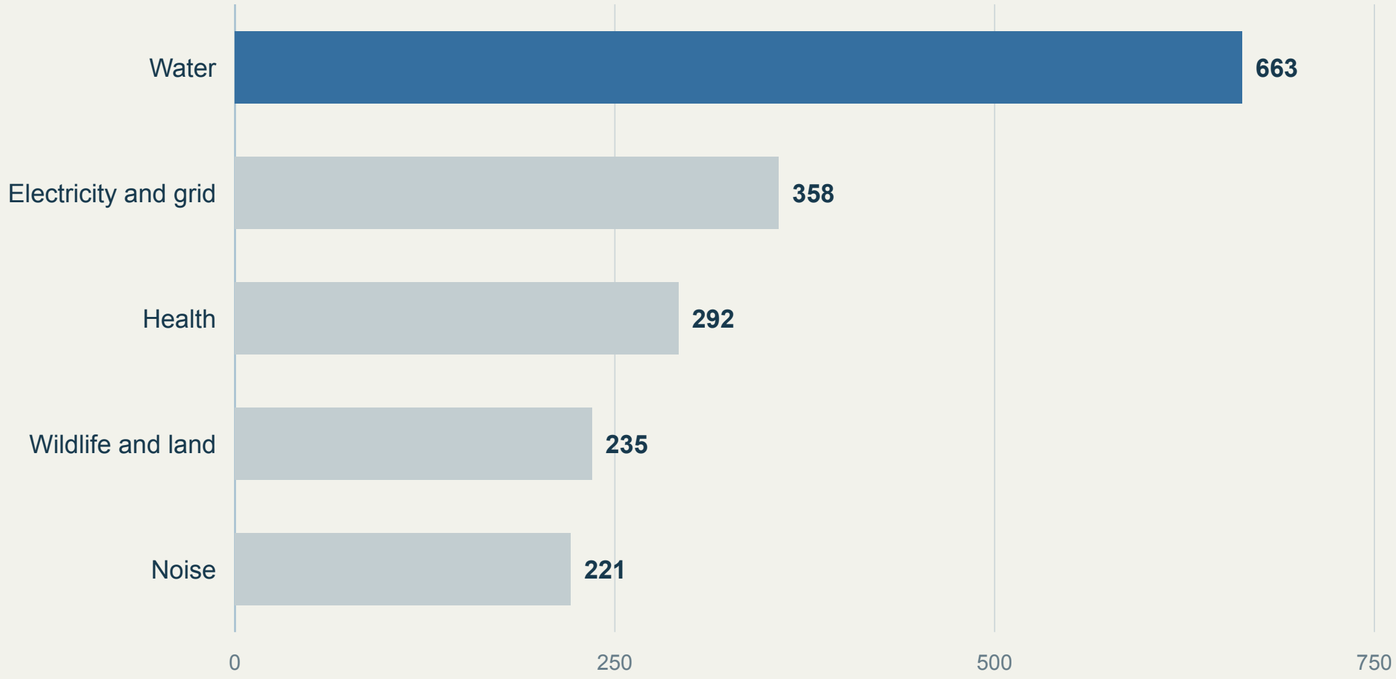
Sources: LBNL, 2024 United States Data Center Energy Usage Report (December 2024); Google 2026 Environmental Report (consumption 2021–2025); Microsoft 2026 Environmental Data Fact Sheet (FY20–FY25). Indexes are Planetary PS calculations. Water-stress share: WRI (February 2026), citing Bloomberg. National totals do not describe any one facility.

Community reports make water, health, and everyday impacts hard to overlook

The tracker offers leads for local investigation; it is not a representative survey or a verified inventory of harms.

Most frequent concern categories in the Brockovich tracker

Keyword matches; categories overlap; snapshot generated October 4, 2026



USE AS A SIGNAL

Ask who gains and who carries the burden

- Compare jobs and tax revenue with incentives, public infrastructure costs, and local exposure
- Publish assumptions and terms

ABOUT THE DATA

Self-selected reports, including on battery storage projects. They do not estimate how all residents feel.

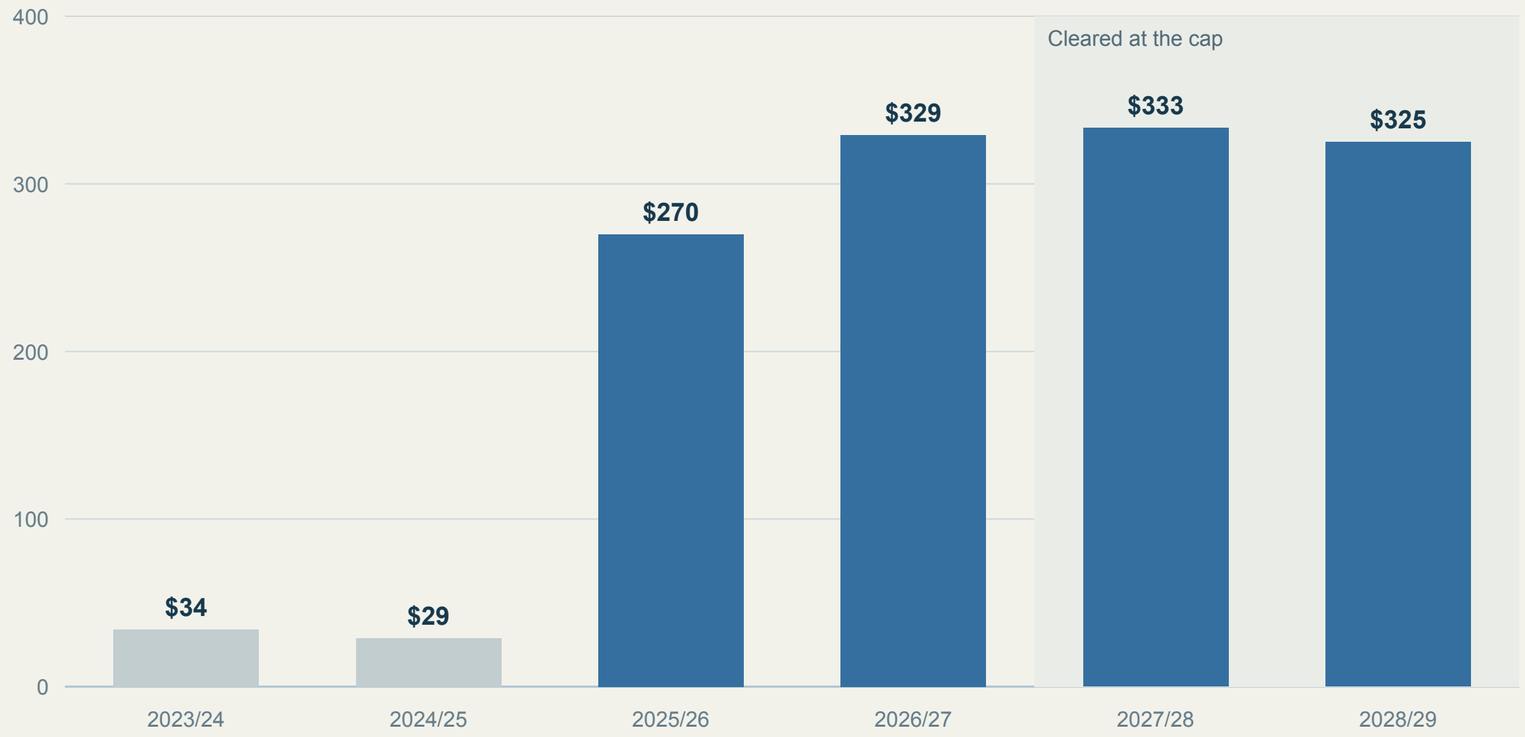
Sources: Brockovich tracker, community statistics and FAQ; WRI (February 2026). Counts are used because the eligible denominator for published concern percentages is not disclosed.

Capacity prices in the largest US grid have risen more than tenfold

PJM's capacity price went from $28.92 per MW-day for 2024/25 to $325 for 2028/29.

PJM capacity auction clearing price, by delivery year

USD per MW-day (RTO, UCAP)



WHO PAYS

Data-center load drives much of the capacity cost

- The market monitor attributes $6.3 billion of the 2028/29 auction's $16.4 billion, 38%, to data-center load
- These are wholesale charges; household bills depend on each utility

THE COUNTERCASE

EPRI finds new load can lower average rates where its incremental cost is low and tariffs assign it.

Sources: PJM, 2028/2029 Base Residual Auction Report (July 14, 2026); Monitoring Analytics, report to the Members Committee (July 28, 2026), pp. 4, 13–14; EPRI, Win-Win Watts (January 2026).

In 2026, states started deciding who pays and who waits

Tariffs, permit pauses, and curtailment rules arrived within months, mostly from governors and regulators.

WHO PAYS

Large-load tariffs

Separate rate classes and tariff minimums put grid costs on the loads that cause them, at 20 to 100 MW. A Senate bill would do the same for transmission.

WHO WAITS

Pauses and audits

New York and Texas slowed permits and connections while they study the queue. Texas alone has 474 GW of requests, about 90% from data centers.

WHO DECIDES

Local approval and disclosure

Virginia’s governor barred state agencies from data-center NDAs and will ask lawmakers in 2027 to require local approval above 25 MW.

SELECTED 2026 ACTIONS

Apr 24	Maine’s governor vetoes a statewide moratorium bill.
Apr 30	Pennsylvania’s PUC adopts a model tariff for loads over 50 MW.
May 7	Florida enacts SB 484: tariff minimums and limits on water permits.
Jul 14	New York pauses new state permits for hyperscale data centers.
Aug 3	Texas orders an audit before more data centers can connect.
Aug 26	Delaware creates a separate rate class for loads of 20 MW or more.
Sep 18	Virginia bars state agencies from data-center NDAs.
Sep 21	Texas halts state environmental permits for data centers.
Oct 1	**Pennsylvania proposes curtailing large new loads first in emergencies.**

Sources: releases from Maine, Pennsylvania, Florida, New York, Texas (permits), Virginia, and Delaware; NCSL; Senate ENR Committee. Not an exhaustive list.

FOR DECISION-MAKERS

Five questions to ask before a data center is approved

DEMAND

Is the load real, and who pays if it is late?

Ask for contracts and a phased ramp. Minimum bills and long contracts keep costs on the project if it shrinks or leaves.

426–649

TWh, the range of 2030 US estimates (IEA, LBNL)

POWER

Does it bring new clean supply?

In the same grid and, ideally, the same hours, with flexibility at peaks. Otherwise the marginal plant serves it, and in much of the US that is gas.

318 Mt

of CO_2 a year if proposed data-center gas plants are built (BloombergNEF)

PLACE

Which basin, and which power plant?

Cooling draws on the local watershed; the power plants draw on others. Ask for both, and for the stress level of each.

12×

water used through electricity versus on site, US total, 2023

PLACE

What does the site displace?

Campuses keep getting bigger. Check habitat, wetlands, and farmland before zoning, not after.

224 acres

average campus in 2024, up 144% since 2022 (WRI)

COMMUNITY

Are the terms public, and has the community had a real say before approval?

Planetary PS synthesis; figures as cited on earlier slides. WRI (February 2026), citing Cushman & Wakefield; minimum bills and long contracts: EPRI (January 2026).

SOURCES AND METHOD

Every chart in this readout is drawn from published data

DEMAND
Slides 3–5

IEA, LBNL, EPRI, GEM, and Currence

IEA, Key Questions on Energy and AI (April 2026); LBNL 2025 Update; EPRI, Powering Intelligence 2026; Global Energy Monitor; IEA capex chart; Currence.

POWER
Slides 6–10

IEA, Bloomberg, company reports, EIA, Ember, Duke, and field trials

IEA (2025); BloombergNEF; Microsoft, Google, Amazon, Meta; EIA-860M; Ember; Duke; Colangelo et al..

PLACE
Slides 11–15

WRI, LBNL, trackers, PJM, EPRI, and the states

WRI; LBNL 2024 report; Brockovich map; PJM; PJM market monitor; EPRI, Win-Win Watts; state releases; NCSL.

METHOD
Read with care

Where we calculated

Indexes, percentage changes, IEA source-level steps, and EIA sums by technology are Planetary PS calculations from published figures. Estimates, scenarios, pipelines, and trials are labeled as such.